

Chapter 5\movie1_FuncAnimation.py

```
1  """
2  As movie1_mpl1.py, but using the newer FuncAnimation
3  set-up for animations.
4  """
5  import numpy as np
6  import matplotlib.pyplot as plt
7  import matplotlib.animation as animation
8  import time, glob, os
9
10 for filename in glob.glob('tmp*.png'):
11     os.remove(filename)
12
13
14 def f(x, m, s):
15     return (1.0/(np.sqrt(2*np.pi)*s))*np.exp(-0.5*((x-m)/s)**2)
16
17 m = 0
18 s_max = 2
19 s_min = 0.2
20 x = np.linspace(m - 3*s_max, m + 3*s_max, 1000)      # Set x-axis limit for plot
21 s_values = np.linspace(s_max, s_min, 30)             # Array of s values
22 # f is max for x=m; smaller s gives larger max value
23 max_f = f(m, m, s_min)
24
25 # Make a first plot (save the lines objects returned from plt.plot)
26 fig = plt.figure()
27 plt.axis([x[0], x[-1], -0.1, max_f])
28 lines = plt.plot([], [])
29 plt.xlabel('x')
30 plt.ylabel('f')
31
32 # Function to return the background plot in the animation
33 def init():
34     lines[0].set_data([], []) # empty plot
35     return lines
36
37 # Function to return a frame in the movie
38 def frame(args):
39     frame_no, s, x, lines = args
40     y = f(x, m, s)
41     lines[0].set_data(x, y)
42     # Does not work: lines[0].set_label('s=%4.2f' % s)
43     # Does not work: plt.legend(['s=%4.2f' % s])
44     # Does not work: plt.savefig('tmp_%04d.png' % frame_no)
45     return lines
46
47 # Construct list of all arguments to frame function
48 # (each call sends frame number, s value, x array, and lines list)
```

```
49 all_args = [(frame_no, s, x, lines)
50              for frame_no, s in enumerate(s_values)]
51
52 # Run the animation
53 anim = animation.FuncAnimation(
54     fig, frame, all_args, interval=150, init_func=init, blit=True)
55
56 # Make movie file in MP4 format
57 anim.save('movie1.mp4', fps=5)
58 plt.show()
59
60
```